When Memory Is Observable: Control Bounds and Temporal-Coarse-Graining Obstructions

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Abstract

An increase of distinguishability in an averaged qubit evolution can be dynamically real yet difficult to identify from measured records. For a qubit coupled along a fixed axis to a symmetric classical telegraph process, we derive, in the fast-switching regime $\gamma \geq 4v$ $(0 < v/\gamma \leq 1/4)$, an all-control ceiling of order $v^2 D^2/\gamma^4$ on each endpoint increase of squared trace distance when transverse drive strength is bounded. An explicit constant-drive construction attains that order. We then consider unresolved timing: without a common bound on control speed, one continuous unitary trajectory can approximate the entire time-averaged path of any continuous unital qubit channel. This gives a uniform finite-sample indistinguishability result for independently reset, single-time experiments. A separate, externally calibrated set of speed-and-clock assumptions restores a nonempty distinguishability region. For a specified eight-setting, point-time experiment, information bounds make a fourth-power trial cost necessary. Independent readout flips at a fixed nonzero rate raise this necessary order to the eighth power, even with exactly known mean bias. Standard quantum-channel theory and classical noise are assumed. No physical bath, instrument precision, experimental confidence or practicable shot count is demonstrated.

1 A dynamics claim is not yet a measurement claim

The relevant comparison is between two explanations for an observed change in distinguishability: an averaged evolution with memory, and an ordinary fine-time evolution viewed through an unresolved observation clock. The comparison class, reset rule and calibration are parts of the theorem. Trace-distance backflow is the established Breuer–Laine–Piilo witness [1]; completely positive divisibility and correlation-based criteria are distinct notions [2, 3]. This paper bounds individual returns of squared trace distance, not the complete positive-variation measure, and does not identify that witness with full multitime process memory. Classical telegraph and colored noise already produce memory effects [4, 5, 6]. The contribution sought here is the bounded-drive fast-switching estimate and its acquisition constraints, not memory from classical noise itself.

Unital-qubit canonical forms and decompositions into unitary channels are established [7, 8]. We use them rather than claim a new channel classification. The candidate contributions are an explicit bounded-drive visibility law, a time-path construction with a uniform error estimate, and their operational separation. Comparison with the broader control, relaxation and statistical-testing literature has not been attempted here; no priority is claimed for these bounds. Exact telegraph-noise dynamics and pulse control are established [10]; the present drive contract instead excludes

instantaneous pulses of unbounded transverse strength. The time-path construction is a quantitative application of relaxed-control/chattering ideas [11], not a claim to have discovered rapid-control approximation. Its observation model is independently reset single-time experiments, not a general multitime quantum-process tomography theorem.

2 Classical telegraph model and conventions

Let a symmetric two-state process switch at rate $\gamma > 0$, and couple to a qubit with Hamiltonian contribution $\xi(t)v\sigma_z$, $\xi = \pm 1$. A prescribed system Hamiltonian is $b(t) \cdot \sigma$. For half the Bloch-vector difference of any two preparations write m for the mean and h for the signed conditional difference. In time $s = \gamma t$,

$$m' = \Omega m + 2\epsilon J h, \qquad h' = \Omega h + 2\epsilon J m - 2h, \quad Jx = e_z \times x, \quad \epsilon = v/\gamma. \tag{1}$$

The independent balanced initial noise state gives $h(0) = 0$ and $\|m(0)\| \le 1$. Require

$$\|[\Omega(s), J]\| \le 2\delta, \qquad \delta = D/\gamma,$$

equivalently $\sqrt{b_x^2 + b_y^2} \le D$ in the physical Hamiltonian. The longitudinal field is locally bounded but need not obey the same cap. Controls are piecewise continuous and prescribed, not feedback using the hidden fluctuator. Neither a bandwidth cap nor an instantaneous-pulse model is implied.

The trace distance of the two mean states is $\|m\|$. Skew symmetry gives the exact balance

$$(\|m\|^2 + \|h\|^2)' = -4\|h\|^2. \tag{2}$$

This is a distinguishability identity, not a thermodynamic-energy law.

3 A uniform endpoint-return ceiling

Theorem 3.1. *For $0 < \epsilon \le 1/4$ and every admitted control and preparation,*

$$\|m(s_2)\|^2 - \|m(s_1)\|^2 \le \frac{16\epsilon^2\delta^2}{(1 + \sqrt{1 - 4\epsilon^2})^4} < 2\epsilon^2\delta^2 \quad (s_2 \ge s_1).$$

The final strict inequality is non-strict in a degenerate zero-coupling or zero-transverse-drive case.

Proof. Put $\beta = \sqrt{1 - 4\epsilon^2}$, $\lambda = 1 + \beta$ and

$$k(s) = \frac{2\epsilon(1 - \mathrm{e}^{-2\beta s})}{1 + \beta - (1 - \beta)\mathrm{e}^{-2\beta s}}.$$

Then $k' = 2\epsilon - 2k + 2\epsilon k^2$, $k(0) = 0$, and $0 \le k \le 2\epsilon/\lambda$. The residual $r = h - kJm$ obeys

$$r' = (\Omega - 2I - 2\epsilon k J^2)r + k[\Omega, J]m.$$

Its homogeneous symmetric part is at most $-\lambda I$. The balance (2) gives $\|m\| \le 1$, hence

$$\|r\| \le 2\delta \int_0^s \mathrm{e}^{-\lambda(s-u)} k(u)\,\mathrm{d}u \le 4\epsilon\delta/\lambda^2 =: R.$$

Let $f = \|m\|^2$, $g = \|h\|^2$ and $x = Jm$. Direct calculation gives $f' = -4\epsilon(k\|x\|^2 + x \cdot r)$. Whenever $f' > 0$,

$$g = \|r\|^2 + 2k(k\|x\|^2 + x \cdot r) - k^2\|x\|^2 \le R^2.$$

If a positive derivative occurs between two endpoints, let τ be the infimum of such times. Before τ, f is nonincreasing; continuity gives $g(\tau) \le R^2$. Nonincrease of $f + g$ then gives $f(s_2) - f(s_1) \le g(\tau) - g(s_2) \le R^2$. If there is no positive derivative, the conclusion is immediate. Finally $\lambda > 7/4$ and $16/(7/4)^4 < 2$. □

The theorem bounds each endpoint increase. It does not bound the sum of all positive variations over an arbitrarily long record. The fast-switching assumption is essential to the stated ceiling. With no drive and an initially transverse vector, eliminating h gives $m'' + 2m' + 4\epsilon^2 m = 0$. For $\epsilon > 1/2$, its scalar factor is $\mathrm{e}^{-s}(\cos\omega s + \sin\omega s/\omega)$, $\omega = \sqrt{4\epsilon^2 - 1}$, and it has distinguishability revivals even though $D = 0$. The present ceiling makes no claim in that regime.

4 A matching-order construction

Proposition 4.1. *For $0 < \epsilon, \delta \le 1/32$, use a constant drive $D\sigma_x$ and*

$$m(0) = \frac{(0, 2\delta, 1 - \delta^2)}{1 + \delta^2}.$$

Between $s = 1$ and $s = 5/4$ the squared-distance return Δ satisfies

$$\frac{\epsilon^2\delta^2}{32} < \Delta < 2\epsilon^2\delta^2.$$

Proof. The invariant subsystem for $H = h_x/(\epsilon\delta)$, $Y = m_y/\delta$, $Z = m_z$ is

$$H' = -2H - 2Y, \quad Y' = 2\epsilon^2 H - 2Z, \quad Z' = 2\delta^2 Y.$$

The rescaled zero-parameter reference is $Z_0 = 1$, $Y_0 = 2 - 2s$, $H_0 = 2s - 3 + 3\mathrm{e}^{-2s}$. The exact return is $4\epsilon^2\delta^2 \int_1^{5/4} YH \,\mathrm{d}s$, whose reference coefficient is

$$c_0 = \frac{1}{6} - 6\mathrm{e}^{-2} + 9\mathrm{e}^{-5/2} > \frac{9}{100}.$$

The integral equations on $[0, 5/4]$ give $\sup|Y| < 5$, $|Z - 1| \le (29/2)\delta^2$, and

$$|Y - Y_0|,\ |H - H_0| \le E := \frac{25}{2}\epsilon^2 + \frac{153}{4}\delta^2 \le \frac{203}{4096}.$$

On the integration interval, $|Y_0| \le 1/2$ and $|H_0| < 3/5$. Consequently the coefficient error is at most $11E/10 + E^2$. The rational inequality $9/100 - 11(203/4096)/10 - (203/4096)^2 > 1/32$ proves the lower bound. The upper bound is Theorem 3.1. □

At fixed v, D, the order is γ^{-4}. The preparation depends on D/γ and the observation window scales as $1/\gamma$. This is not a fixed-preparation, fixed-clock prediction.

5 A whole-path timing obstruction

Write $d(\Phi,\Psi)=\frac{1}{2}\|\Phi-\Psi\|_\diamond$. It is at most one for channels and bounds all single-outcome probability differences, including ancilla-assisted ones.

Every unital qubit channel is an equal average of four unitary channels [8]. For a diagonal canonical channel with Bell–Choi probabilities $p_0,\dots,p_3$, one construction is

$$U_k=\sqrt{p_0}I-\mathrm{i}\sum_{a=1}^{3}s_{ka}\sqrt{p_a}\sigma_a,$$

where the sign rows are $(+++),(+--),(-+-),(--+)$. Averaging kills the cross terms. Outer unitary conjugations restore the general channel.

Theorem 5.1 (Time-averaged closure)**.** *Let $\mathcal{C}(s)$ be a diamond-continuous unital-qubit channel path on $[0,T]$, with $0<w<T/2$, and define*

$$\overline{\mathcal{C}}(t)=\frac{1}{2w}\int_{t-w}^{t+w}\mathcal{C}(s)\,\mathrm{d}s,\quad w\le t\le T-w.$$

For every $\varepsilon>0$ there is one continuous, piecewise-smooth $U(s)\in SU(2)$ with $U(0)=I$ whose averaged conjugation channel is uniformly within ε of $\overline{\mathcal{C}}$. Each chosen path has finite-speed pieces, but no common speed cap as $\varepsilon\downarrow 0$ is asserted.

Proof. Divide time into equal cells of length ℓ. Freeze the target at each left endpoint and dwell equally at its four representing unitaries. If $\omega(\ell)$ is the target modulus of continuity, freezing costs at most $\omega(\ell)$. Full cells have the exact same average; the two possible partial cells cost at most ℓ/w.

Replace each jump by an $SU(2)$ arc in the first $\tau\le\ell/4$ of the next subcell, including an initial arc from I. At most $8w/\ell+2$ arc intervals intersect an averaging window. Their changed fraction is at most $4\tau/\ell+\tau/w$. Therefore

$$\sup_t d(\overline{\mathcal{U}}(t),\overline{\mathcal{C}}(t))\le\omega(\ell)+\ell/w+4\tau/\ell+\tau/w.$$

Choose $\omega(\ell)\le\varepsilon/4$, $\ell\le\varepsilon w/4$ and $\tau\le\varepsilon\ell/16$, reducing ε to at most one if needed. The bound is at most $49\varepsilon/64$. The connecting arcs have finite generators for each positive τ. □

This approximates an *already averaged* target path. It does not realize arbitrary incompatible maps at overlapping windows, nor does it cover nonunital channels or higher-dimensional quantum Birkhoff problems.

Corollary 5.2 (Uniform finite-sample obstruction)**.** *Assume independent reset trials and fresh independent uniform clock draws; only nominal times and outcomes are recorded. For any finite number N of single-time channel uses, a test with size at most α for every admitted fine-time unitary path without a common speed cap has rejection probability at most α on the averaged target.*

Proof. The uniform channel approximation gives a record-law total-variation distance at most $N\varepsilon$ by the hybrid argument, including adaptive input/effect choices and single-time ancillas. The rejection probability is at most $\alpha+N\varepsilon$ for every $\varepsilon>0$. □

The approximating path may change with ε. This is not one fixed finite-speed null with identical laws at arbitrarily large sample sizes. Within-trial multitime interventions and time-resolved records are outside the experiment.

6 What external calibration restores

Under an independently justified pair-state speed cap of 4, the special target $v = D = 1$, $\gamma \geq 32$ admits a sufficient clock-and-error region

$$w \leq \frac{1}{256\gamma^2}, \qquad \eta_{\text{other}} \leq \frac{\gamma^{-4}}{2400}.$$

The squared distinguishability witness from three first-time orthogonal axes and one free-rotated final axis has an all-centres ideal margin greater than $\gamma^{-4}/500$. The null clock allowance is $16w^2/3 \leq \gamma^{-4}/12288$, giving a clock-aware margin greater than $\gamma^{-4}/600$. The target's probability curvature bounds its centered jitter error by $7w^2/6$; that error and η_{other} stay below the required total $\gamma^{-4}/1200$ budget. These inequalities are a conditional acquisition region, not achieved instrument performance. Their detailed calibration proof is included in the supplement as the bounded-drive and tomography-clock proofs. They use a different set of assumptions from the unrestricted-speed obstruction.

The operational prediction, within $\gamma \geq 4v$, is that increasing the switching rate at fixed coupling and drive makes the best endpoint visibility scale no better than the fourth inverse power, while a tailored admitted preparation attains that order. More samples cannot compensate for an unrestricted clock/control null class; an external bound must first narrow that class.

7 Statistical cost and the readout trade-off

We now distinguish a probability-error allowance from the number of trials required to attain it. Throughout this section $v = D = 1$, $\gamma \geq 32$, $\delta = \gamma^{-1}$ and $\sigma = \gamma^{-4}$. At $s = 1$ measure both opposed preparations in the orthonormal triad

$$f_1, \quad e_x, \quad (0, f_{1z}, -f_{1y}),$$

and at $s = 5/4$ along f_2, where f_i is the freely rotated initial vector at the corresponding time. All axes and counts are fixed before acquisition. This gives eight binary settings, with pair gains g_1, g_2, g_3, g_4 and witness $W = g_4^2 - g_1^2 - g_2^2 - g_3^2$.

The following control and axis-error estimates for the target are proved in the supplement [12]:

$$g_2 = 0, \qquad \sigma/48 < W < 2\sigma, \qquad q_1, q_4 \leq C_0\sigma, \quad C_0 = 7109689/4194304 < 7/4, \tag{3}$$

$$|g_3| \leq (85/1024)\sqrt{\sigma}, \qquad \|m_1 - f_1\| \leq A\delta^3, \quad A = 85/32. \tag{4}$$

Here $q_i = (1 - g_i)/2$ are the aligned failures for either preparation of this unital target. The detector and the null do not assume unitality. These estimates are not consequences of a finite numerical scan. Their proofs are in Section 6 of *A bounded-strength endpoint-return law* and Section 7 of *Time-averaged tomography, closure, and an admissible region.* The separate *Finite-sample rare-event return witness* takes these as inputs for the count argument. The supplement README maps these titles to the supplied files. Finite interval controls do not replace those proofs.

7.1 A sound finite-count rule

Let $a = k/n$ be the observed failure fraction in an aligned setting, and let $\vartheta > 0$ be the confidence parameter. The data-derived radius

$$R(a, n, \vartheta) = \sqrt{2a\vartheta/n} + 4\vartheta/n$$

covers the true Bernoulli mean with error probability at most $2e^{-\vartheta}$, for every mean, including nonrare nulls. Indeed Bernstein's bound gives $|a-q| \le \sqrt{2q\vartheta/n} + 2\vartheta/(3n)$. For $q > a$, solving the quadratic gives

$$q - a \le \frac{5\vartheta}{3n} + \sqrt{\frac{2a\vartheta}{n} + \frac{7\vartheta^2}{3n^2}} < R(a, n, \vartheta);$$

for $q \le a$ the result is immediate. In particular, zero observed failures still leave radius $4\vartheta/n$.

At an aligned axis use $\widehat{g} = 1 - a_+ - a_-$ and $e = R_+ + R_- + 2\beta$, where β bounds each probability's nonclock systematic error. At a transverse axis use the difference of positive-outcome frequencies and $e = \sqrt{\vartheta/(2n_+)} + \sqrt{\vartheta/(2n_-)} + 2\beta$. Reject only if

$$\max(0, |\widehat{g}_4| - e_4)^2 - \sum_{j=1}^{3} \min(1, |\widehat{g}_j| + e_j)^2 > 16w^2/3. \tag{5}$$

IID reset trials within each setting and the prescribed clock law imply simultaneous coverage except with probability $16e^{-\vartheta}$. On coverage, the expression is a lower bound on the clock-averaged witness; the qualified contractive null cannot exceed $16w^2/3$. Soundness uses no rare-event property of the null.

With $w \le 1/(256\gamma^2)$ and $\beta \le \sigma/100000$, the detailed target power estimate in the supplied confidence/count proof gives the sufficient count

$$N_{\text{suff}} = 4\lceil 2^{22}\vartheta\gamma^4 \rceil + 4\lceil 2^{16}\vartheta\gamma^4 \rceil \tag{6}$$

for both errors at most $16e^{-\vartheta}$, $\vartheta \ge 3$. The first four counts use aligned failures; the other four use transverse positive outcomes. This is a conservative sufficient guarantee, not an optimal test. It includes both displacement of the empirical centre and the confidence radius, and does not divide systematic error by sample size.

7.2 Why fourth-power cost cannot be removed at point times

The following lower bound applies to independent single-time trials chosen from these eight settings at $w = 0$. It is not a bound on arbitrary multitime quantum experiments. The null includes CP-divisible rotations and isotropic depolarization, with pair-state speed at most four in physical time.

Theorem 7.1 (Point-time information obstruction)**.** *If both errors of such a test are at most* $16e^{-\vartheta}$*, where* $\vartheta > \log 64$*, then its total trial count satisfies*

$$N > 2(\vartheta - \log 64)(1 - 6\sigma)\gamma^4.$$

Proof. Put $C = 7/4$, $b_* = 1 - 2C\sigma > 1/2$ and $r_1 = \sqrt{g_1^2 + g_3^2} \ge g_1 \ge b_*$. A null can match the first-time vector, then rotate at fixed norm r_1 to the final measurement axis. Its only differing setting probabilities have shift

$$d = (g_4 - r_1)/2 = \frac{W}{2(g_4 + r_1)} < \frac{\sigma}{2b_*}.$$

To check admissibility, interpolate the norm from one to r_1 by isotropic depolarization and rotate along the shorter arc from the initial vector to m_1/r_1; then rotate at fixed norm to f_2. Here $b_*^2 \le 1 - A^2\delta^6$, so the vector-error bound and $\arcsin u \le u/\sqrt{1-u^2}$ give angle bounds $2\delta + A\delta^3/b_*$

and $\delta/2 + A\delta^3/b_*$. Linear schedules have pair speeds at most $2 + A\delta^2/b_* + 2C\delta^3$ and $2 + 4A\delta^2/b_*$, both below three. The path is therefore an admitted CP-divisible null, not merely a list of unrelated endpoints.

For a changed failure probability q and $q' = q + d$, the elementary Bernoulli relative-entropy bound gives

$$\begin{aligned} D_{\mathrm{KL}}(q\|q') &\le \frac{d^2}{q'(1-q')} \le \frac{d}{1-q'} \\ &< \frac{\sigma}{2\{b_*(1-C\sigma) - \sigma/2\}} \le \frac{\sigma}{2(1-6\sigma)}. \end{aligned}$$

The final inequality uses $3C + 1/2 = 23/4 < 6$. The Bretagnolle–Huber testing inequality gives $\alpha + \beta_{\mathrm{II}} \ge \frac{1}{2}\exp(-D_{\mathrm{KL}}(P\|Q))$ [9, Cor. 4]. Since the total information is at most N times the largest per-trial divergence and $\alpha + \beta_{\mathrm{II}} \le 32e^{-\vartheta}$, the stated result follows. □

The point-time case is a necessary obstruction for any rule uniform over an allowed clock class containing $w = 0$. Constants for a fixed nonzero jitter require the actual averaged target and null probabilities.

7.3 Systematic masking and random detector errors are different

Within a split-null family that increases g_1 by $2d$ and decreases g_4 by $2d$, equal endpoint probability shifts satisfy

$$(g_4 - 2d)^2 = (g_1 + 2d)^2 + g_3^2, \qquad \beta_{\mathrm{split}} = \frac{d}{2} = \frac{W}{8(g_4 + g_1)}.$$

An error allowance β_{split} on each of the two hypotheses admits their common midpoint law. This is a masking example for settingwise adversarial bias, not the global minimax threshold.

If all eight settings instead share one affine readout $p_{\mathrm{obs}} = a_{\mathrm{det}} + b_{\mathrm{det}}p$ with $b_{\mathrm{det}} > 0$, differences of opposed preparations cancel the offset exactly:

$$g_{j,\mathrm{obs}} = b_{\mathrm{det}} g_j, \qquad W_{\mathrm{obs}} = b_{\mathrm{det}}^2 W.$$

This preserves the point-time sign. It neither proves shared gain in an apparatus nor cancels sampling variance. In particular, symmetric independent outcome flips are this same affine readout with $a_{\mathrm{det}} = r$ and $b_{\mathrm{det}} = 1 - 2r$: cancellation of the offset in the mean does not cancel the Bernoulli variance floor below.

Proposition 7.2 (Fixed random readout flips change the necessary order)**.** *Under the hypotheses of Theorem 7.1, suppose each outcome is independently flipped with a known fixed probability $r \in (0, 1/2)$. Writing $\mu = 1 - 2r$, every such test requires*

$$N > \frac{4b_*^2 r(1-r)(\vartheta - \log 64)}{\mu^2}\,\gamma^8.$$

Proof. An ideal failure probability becomes $r + \mu q$, bounded between r and $1 - r$. The same null changes it by μd, so

$$D_{\mathrm{KL,obs}} \le \frac{\mu^2 d^2}{r(1-r)} < \frac{\mu^2\sigma^2}{4b_*^2 r(1-r)}.$$

Apply the same testing inequality. Knowing r exactly does not remove the random flips or their variance floor. □

This proposition is detector-specific. Heralded errors, correlated reference readings, or a different acquisition are not excluded by it. With ideal readout the variance per aligned trial is $O(\gamma^{-4})$, whereas the standard deviation is $O(\gamma^{-2})$; those quantities must not be confused.

Corollary 7.3 (Necessary readout-noise crossover)**.** *Keep the point-time experiment of Theorem 7.1 and fix $\vartheta > \log 64$. At each $\gamma \geq 32$, suppose all outcomes are independently flipped with the same known probability $0 < r_\gamma < 1/2$. Set $H = \vartheta - \log 64$ and $b_* = 1 - 7\gamma^{-4}/2$. Then every such test whose two error probabilities are at most $16e^{-\vartheta}$ has total trial count*

$$N > H \max\left\{2(1-6\gamma^{-4})\gamma^4, \quad 4b_*^2 \frac{r_\gamma(1-r_\gamma)}{(1-2r_\gamma)^2}\gamma^8\right\}.$$

If $r_\gamma \asymp \gamma^{-a}$ with $a \geq 0$ and $r_\gamma \leq 1/2 - \eta$ for some fixed $\eta > 0$, this gives $N = \Omega(\gamma^{\max\{4,8-a\}})$. A trial budget $N = O(\gamma^4)$ therefore requires $r_\gamma = O(\gamma^{-4})$ in this model; this condition alone is not a sufficient protocol.

Proof. The preceding KL inequality is finite-parameter and holds for every $0 < r < 1/2$, so it applies directly at $r = r_\gamma$. Data processing under the independent flip channel also bounds the observed per-trial KL by the clean per-trial KL in Theorem 7.1. Combining the two testing bounds gives the maximum. For $\gamma \geq 32$, the factors $2(1-6\gamma^{-4})$ and $4b_*^2$ lie between one and four, independently of r_γ. Away from $1/2$, the ratio $r(1-r)/(1-2r)^2$ is comparable to r, proving the stated powers. Moreover that ratio is at least r on $[0, 1/2)$, so the second bound forces $r_\gamma = O(\gamma^{-4})$ whenever $N = O(\gamma^4)$. □

At $r = 0$ the clean bound applies. At $r = 1/2$ every observed outcome is a fair coin under both hypotheses, so no finite number of these observations meets the stated error requirement. Rates approaching $1/2$ can impose a stronger obstruction than the power comparison above. The corollary does not establish a matching upper bound or extend to correlated or heralded errors, multitime instruments, or a fixed nonzero exposure.

Numerical scale, not an experimental forecast. For $\gamma = 32$, $\vartheta = 10$, a separately coded six-state generator and split-null calculation at 80 digits give a clean-readout lower bound of approximately 5.798×10^{10} trials and $\beta_{\text{split}} \simeq 5.446 \times 10^{-9}$. Equation (6) gives 178,670,639,513,600 trials. With independent flips at $r = 0.001$, the same numerical null family gives a lower bound near 9.883×10^{13}. These lower bounds are neither exact optima nor achieved tests. The portable Arb certificate independently encloses these rounded values using both the six-dimensional (m, h) and the reduced three-dimensional systems at 256 and 384 bits. It certifies finite controls at $\gamma = 32, 64, 128, 256$, and brackets the best lower bound within the specified split-null family at the quoted readout settings. It is not a global minimax optimization or a proof for all γ by sampling. The separate 80-digit calculation remains a floating-point cross-check. No apparatus, shot rate or physical unit conversion is supplied. A viable instrument must address stable common gain *and* sufficiently rare or independently flagged readout errors.

8 Limitations and next physical test

A physical test must separately qualify state preparation, independent reset, the coupling axis, drive strength, clock distribution, gain/offset and systematic error. The derived trial bounds concern the stated model and detector contracts, not an achieved shot budget or hardware rate. The present construction does not identify a quantum bath. The mathematical obstruction and positive region have different assumptions; neither should be substituted for the other after observing an outcome.

Drafting and verification disclosure

Luqman Mahmood directed the research and is responsible for its claims and final text. OpenAI Codex and Anthropic Claude assisted mathematical work, computation, checking and editing. Statements have different evidence classes: written conditional proofs, computer-assisted certificates, numerical demonstrations, and prospective protocols. These are labelled as such in the text and do not constitute external peer review. No claim of literature priority follows from computational verification. The paper and mathematical data are licensed CC BY 4.0. Software source and executable verification scripts are excluded from that grant, with rights reserved unless a file explicitly states otherwise.

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